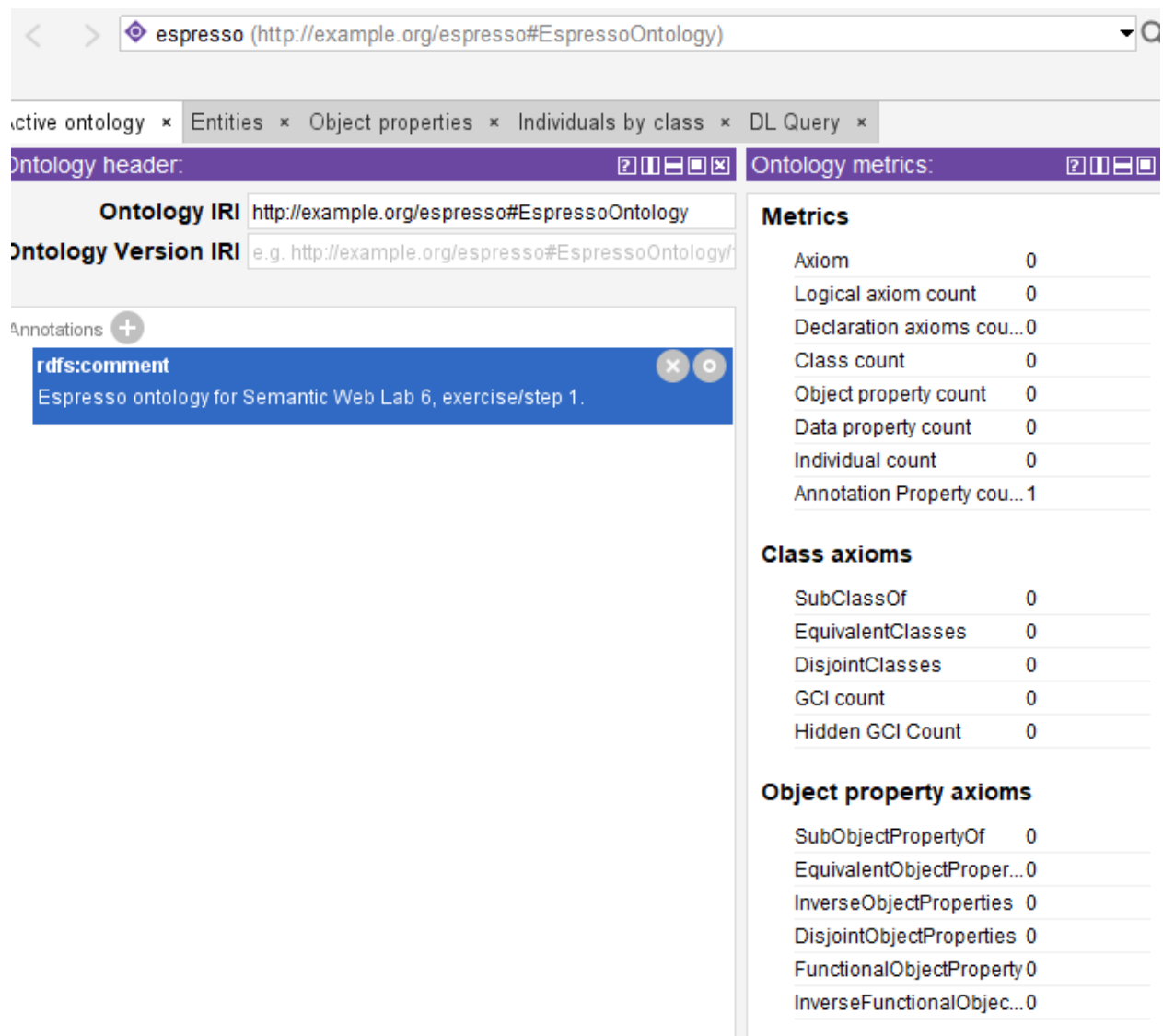


Semantic Web Lab 6

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Exercise 1



espresso (http://example.org/espresso#EspressoOntology)

active ontology × Entities × Object properties × Individuals by class × DL Query ×

Ontology header: ? ? ? ? ?

Ontology metrics: ? ? ? ? ?

Ontology IRI http://example.org/espresso#EspressoOntology

Ontology Version IRI e.g. http://example.org/espresso#EspressoOntology/

Annotations +

rdfs:comment
Espresso ontology for Semantic Web Lab 6, exercise/step 1.

Metrics

Axiom	0
Logical axiom count	0
Declaration axioms count	0
Class count	0
Object property count	0
Data property count	0
Individual count	0
Annotation Property count	1

Class axioms

SubClassOf	0
EquivalentClasses	0
DisjointClasses	0
GCI count	0
Hidden GCI Count	0

Object property axioms

SubObjectPropertyOf	0
EquivalentObjectProperty	0
InverseObjectProperties	0
DisjointObjectProperties	0
FunctionalObjectProperty	0
InverseFunctionalObjectProperty	0

Exercise 2

< >  espresso (http://example.org/espresso#EspressoOntology)

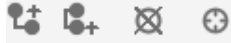
> Espresso

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: Espresso 

 Asserted ▾

owl:Thing

- Espresso
- EspressoBase
- EspressoTopping

Espresso — http://example.org/espresso#Espresso

Annotations Usage

Annotations: Espresso

Annotations 

Description: Espresso

Equivalent To 

SubClass Of 

General class axioms 

SubClass Of (Anonymous Ancestor)

Instances 

Exercise 3

Description: Espresso

Equivalent To

+

SubClass Of

+

General class axioms

+

SubClass Of (Anonymous Ancestor)

Instances

+

Target for Key

+

Disjoint With

+

EspressoTopping

?

@

×

○

EspressoBase

?

@

×

○

Disjoint Union Of

+

Exercise 4

EspressoBase > Café

Active ontology x Entities x Object properties x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: Café

- owl:Thing
 - Espresso
 - EspressoBase
 - Café**
 - Cappuccino
 - EspressoRoast
 - EspressoTopping

Annotations Usage

Annotations: Café

Annotations +

rdfs:label

Café

Description: Café

Equivalent To +

SubClass Of +

- EspressoBase

General class axioms +

SubClass Of (Anonymous Ancestor)

Exercise 5

< > espresso (http://example.org/espresso#EspressoOntology)

> EspressoTopping > CinnamonTopping

Active ontology × Entities × Individuals by class × DL Query ×

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: CinnamonTopping

Annotations Usage

Annotations: CinnamonTopping

Annotations +

Description: CinnamonTopping

Equivalent To +

SubClass Of +

EspressoTopping

owl:Thing

Espresso

EspressoBase

Café

Cappucino

EspressoRoast

EspressoTopping

CinnamonTopping

FoamTopping

WhippedCreamTopping

Asserted

Exercise 6

EspressoTopping CinnamonTopping

Active ontology x Entities x Object properties x Individuals by class x DL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: CinnamonTopping

- owl:Thing
 - Espresso
 - EspressoBase
 - Café
 - Cappucino
 - EspressoRoast
 - EspressoTopping
 - CinnamonTopping
 - FoamTopping
 - WhippedCreamTopping
 - ClassicWhippedCreamTopping
 - SweetenedWhippedCreamTopping
 - VegetableWhippedCreamTopping

Annotations Usage

Annotations: CinnamonTopping

Annotations +

Description: CinnamonTopping

Equivalent To +

SubClass Of +

- EspressoTopping

General class axioms +

Exercise 7

espresso (<http://example.org/espresso#EspressoOntology>)

> hasIngredient

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Datatypes Individuals

Data properties Annotation properties

Classes Object properties

Object property hierarchy: ha: **hasIngredient**

hasIngredient — <http://example.org/espresso#>

Annotations Usage

Annotations: hasIngredient

Annotations

Description: hasIngredient

☐ Functional Equivalent To

Exercise 8

< > espresso (http://example.org/espresso#EspressoOntology)

> hasIngredient

Active ontology * Entities * Object properties * Individuals by class * DL Query *

Object property hierarchy: hasIngredient ? ? ? ? ?

Annotations Usage

Annotations: hasIngredient

Annotations +

owl:topObjectProperty

- hasIngredient
 - hasBase
 - hasTopping

Chara ? ? ? ? ?

Description: hasIngredient

☐ Functional

☐ Inverse functional

☐ Transitive

☐ Symmetric

☐ Asymmetric

☐ Reflexive

Equivalent To +

SubProperty Of +

Inverse Of +

Domains (intersection) +

Exercise 9

The screenshot shows an OWL editor interface. On the left, a tree view displays the hierarchy of the `isIngredientOf` property, which is highlighted with a yellow border. The hierarchy is as follows:

- `owl:topObjectProperty`
 - `hasIngredient`
 - `hasBase`
 - `hasTopping`
 - `isIngredientOf` (highlighted)

The right pane is titled "Annotations" and contains a plus sign icon. Below this, a section titled "Description: isIngredientOf" contains a list of property characteristics, each with an unchecked checkbox:

- ☐ Functional
- ☐ Inverse function
- ☐ Transitive
- ☐ Symmetric
- ☐ Asymmetric

To the right of these checkboxes are three expandable sections, each with a plus sign icon:

- Equivalent To
- SubProperty Of
- Inverse Of

The "Inverse Of" section is currently expanded, showing the property `hasIngredient` as its inverse. A question mark icon is visible at the bottom right of this section.

Exercise 10

isBaseOf

Active ontology × Entities × Object properties × Individuals by class × DL Query ×

Object property hierarchy: isBaseOf

Annotations Usage

Annotations: isBaseOf

Annotations +

owl:topObjectProperty

- hasIngredient
- isBaseOf
- isIngredientOf
- isToppingOf

Chara Description: isBaseOf

- ☐ Functional
- ☐ Inverse functiona
- ☐ Transitive
- ☐ Symmetric
- ☐ Asymmetric
- ☐ Reflexive

Equivalent To +

SubProperty Of +

Inverse Of +

hasBase

Exercise 11

?
⌵
⌶
⌷
⌸

Description: hasIngredient

☐ Functional
 ☐ Inverse function
 ☒ Transitive
 ☐ Symmetric
 ☐ Asymmetric
 ☐ Reflexive
 ☐ Irreflexive

Equivalent To +
 SubProperty Of +
 Inverse Of +

isIngredientOf

?

 Domains (intersection) +
 Ranges (intersection) +
 Disjoint With +
 SuperProperty Of (Chain) +

Basically: X is ingredient of Y, and Y is ingredient of Z, X is ingredient of Z

Exercise 12

?

↑

↓

□

×

Functional

Inverse function

Transitive

Symmetric

Asymmetric

Reflexive

Irreflexive

Description: hasBase

Equivalent To

+

SubProperty Of

+

hasIngredient

Inverse Of

+

isBaseOf

Domains (intersection)

+

Ranges (intersection)

+

Disjoint With

+

SuperProperty Of (Chain)

+

One value only for that property

Exercise 13

Exercise 15

The screenshot displays a web-based ontology editor interface. On the left, a 'Class hierarchy' panel shows a tree structure starting from 'owl:Thing'. Under 'owl:Thing', there is a class 'Espresso', which is the parent of 'EspressoSpecialty'. Below 'Espresso', there is another class 'EspressoBase', which is the parent of 'Café', 'Cappucino', and 'EspressoRoast'. Under 'EspressoBase', there is a class 'EspressoTopping', which is the parent of 'CinnamonTopping', 'FoamTopping', and 'WhippedCreamTopping'. The 'EspressoSpecialty' class is highlighted with a blue box. On the right, a panel for 'EspressoSpecialty' is shown, with tabs for 'Annotations' and 'Usage'. The 'Annotations' tab is active, showing a list of annotations. The first annotation is 'Annotations: EspressoSpecialty'. Below this, there are sections for 'Description: EspressoSpecialty', 'Equivalent To', 'SubClass Of', 'General class axioms', and 'SubClass Of (Anonymous Ancestor)'. The 'SubClass Of' section shows 'Espresso' as a parent class. The 'SubClass Of (Anonymous Ancestor)' section shows 'hasBase some EspressoBase' as an axiom.

Class hierarchy: EspressoSpecialty

- owl:Thing
 - Espresso
 - EspressoSpecialty**
 - EspressoBase
 - Café
 - Cappucino
 - EspressoRoast
 - EspressoTopping
 - CinnamonTopping
 - FoamTopping
 - WhippedCreamTopping

Annotations: EspressoSpecialty

Annotations +

Description: EspressoSpecialty

Equivalent To +

SubClass Of +

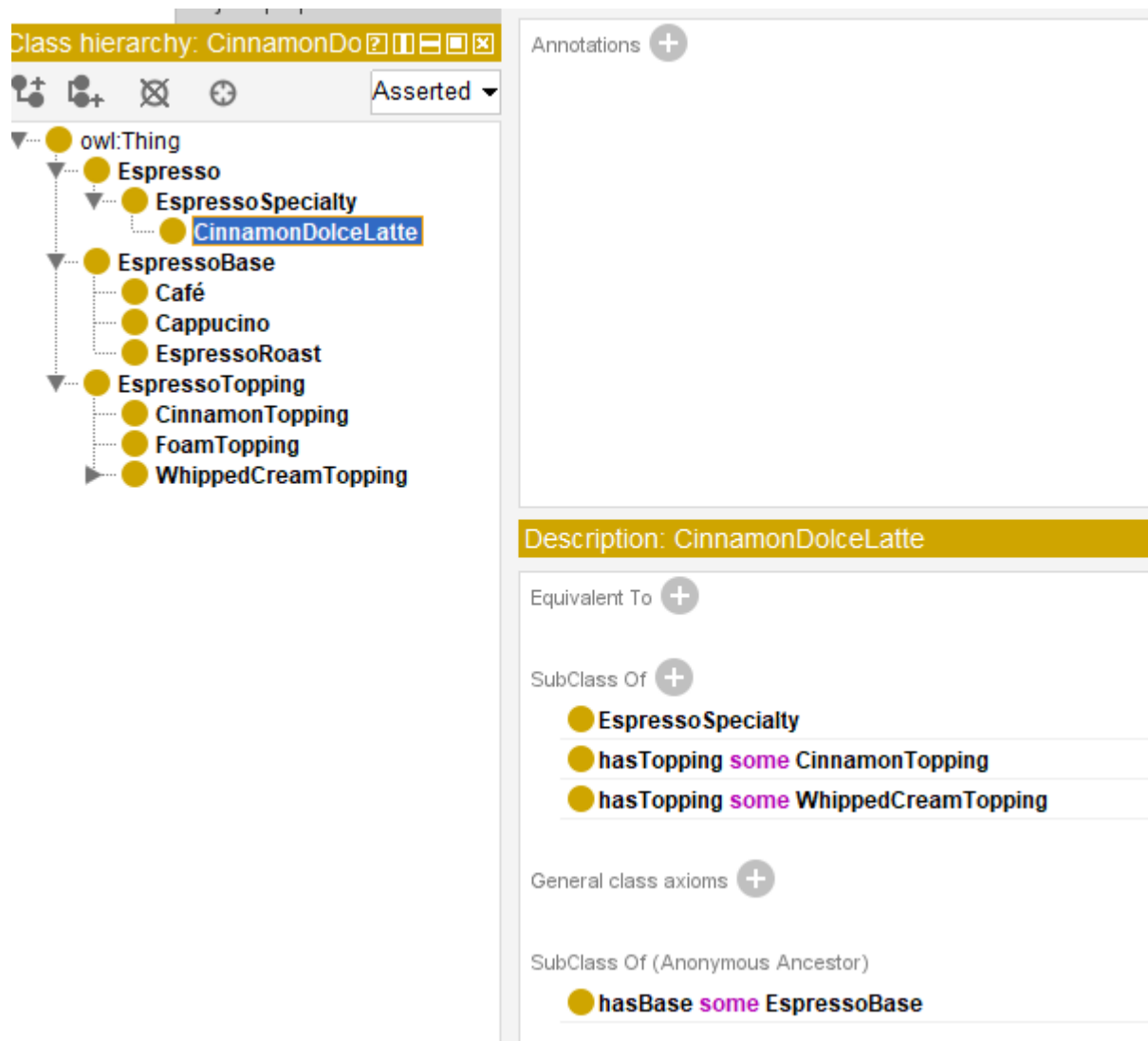
- Espresso

General class axioms +

SubClass Of (Anonymous Ancestor)

- hasBase some EspressoBase

Exercise 16



Exercise 17

Class hierarchy: CinnamonDolceLatte

Asserted

- owl:Thing
 - Espresso
 - EspressoSpecialty
 - CinnamonDolceLatte**
 - EspressoBase
 - EspressoTopping

Annotations: CinnamonDolceLatte

Annotations

rdfs:comment

Espresso with steamed milk and cinnamon dolce flavored syrup.

Description: CinnamonDolceLatte

Equivalent To

SubClass Of

- EspressoSpecialty
- hasTopping some CinnamonTopping
- hasTopping some WhippedCreamTopping

Exercise 18

Class hierarchy: CafeMocha

Asserted

- owl:Thing
 - Espresso
 - EspressoSpecialty
 - CafeMocha**
 - CinnamonDolceLatte
 - EspressoBase
 - EspressoTopping

Description: CafeMocha

Equivalent To

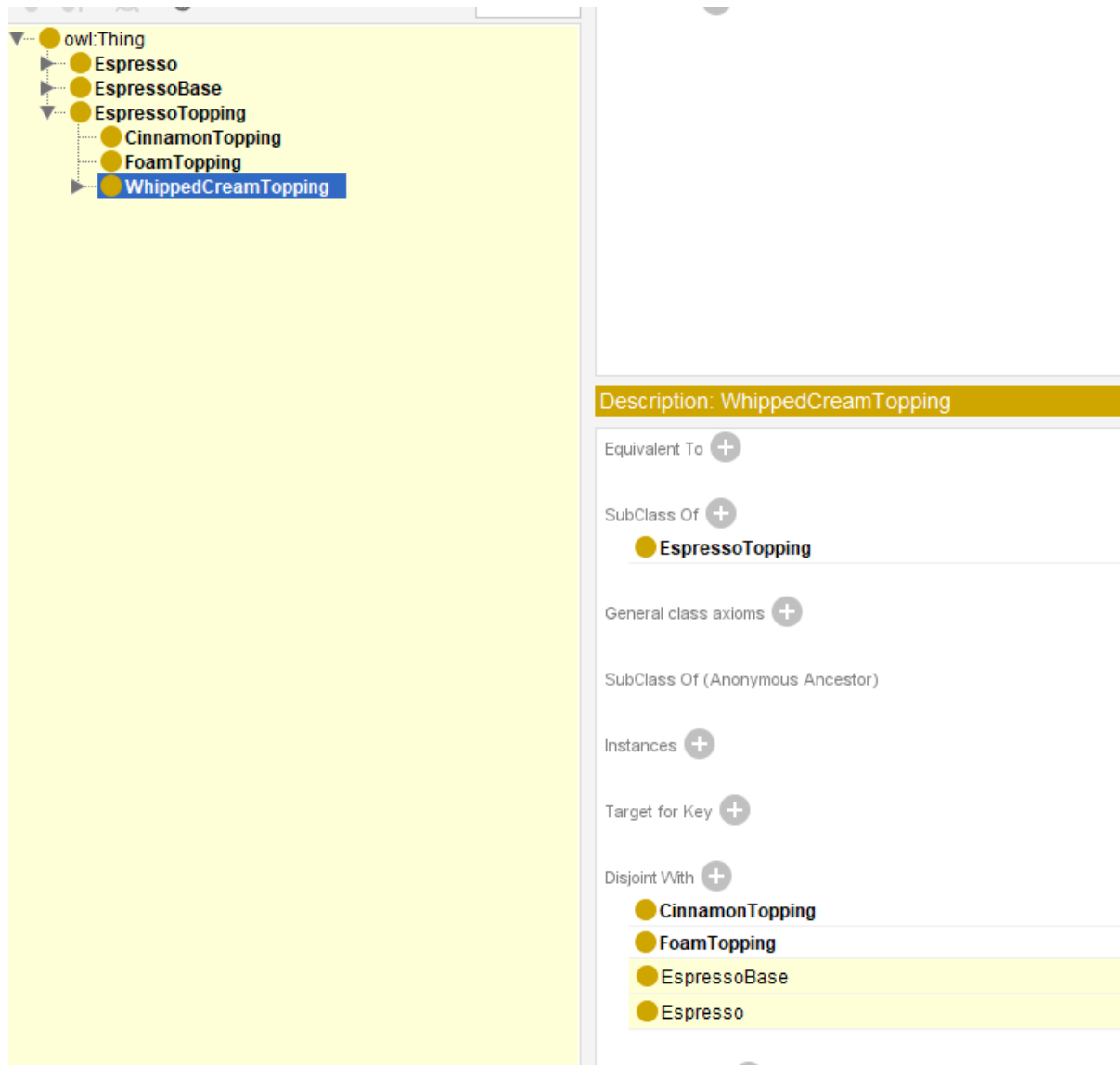
SubClass Of

General class axioms

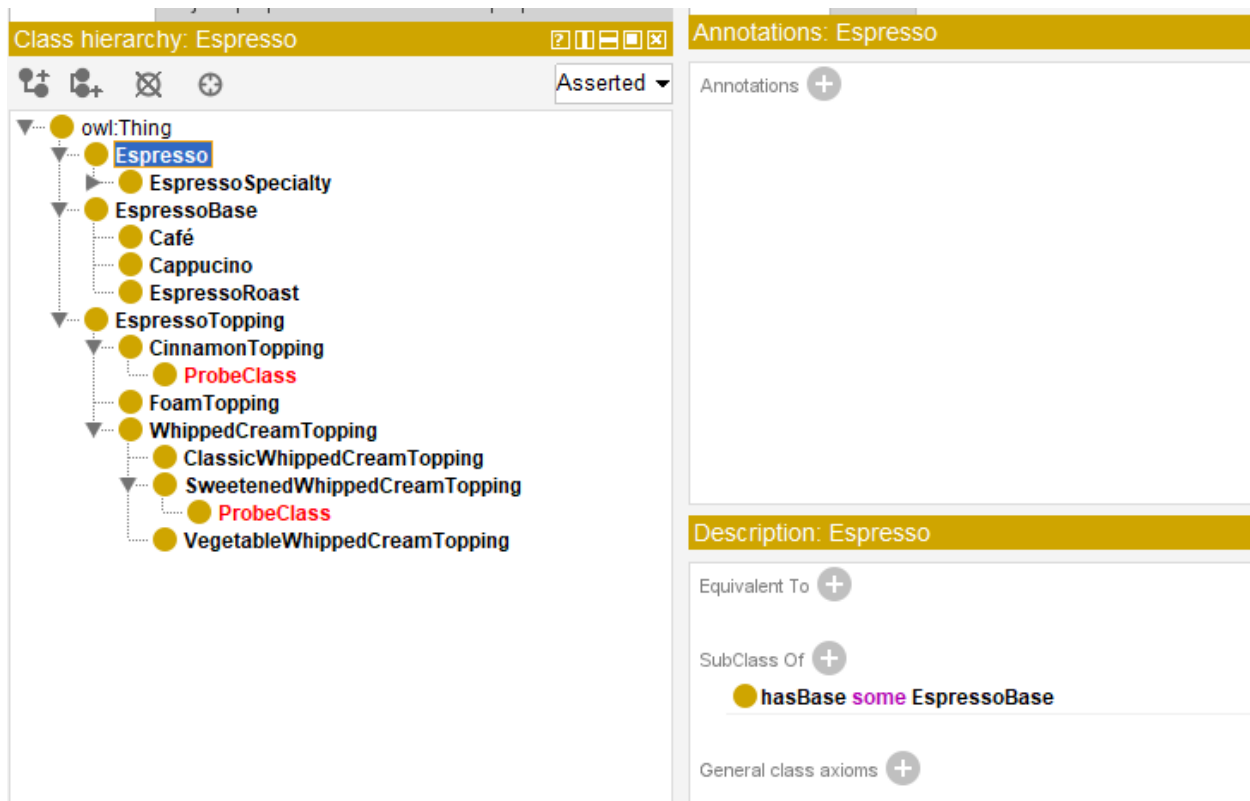
SubClass Of (Anonymous Ancestor)

- EspressoSpecialty
- hasTopping some WhippedCreamTopping
- hasBase some EspressoBase

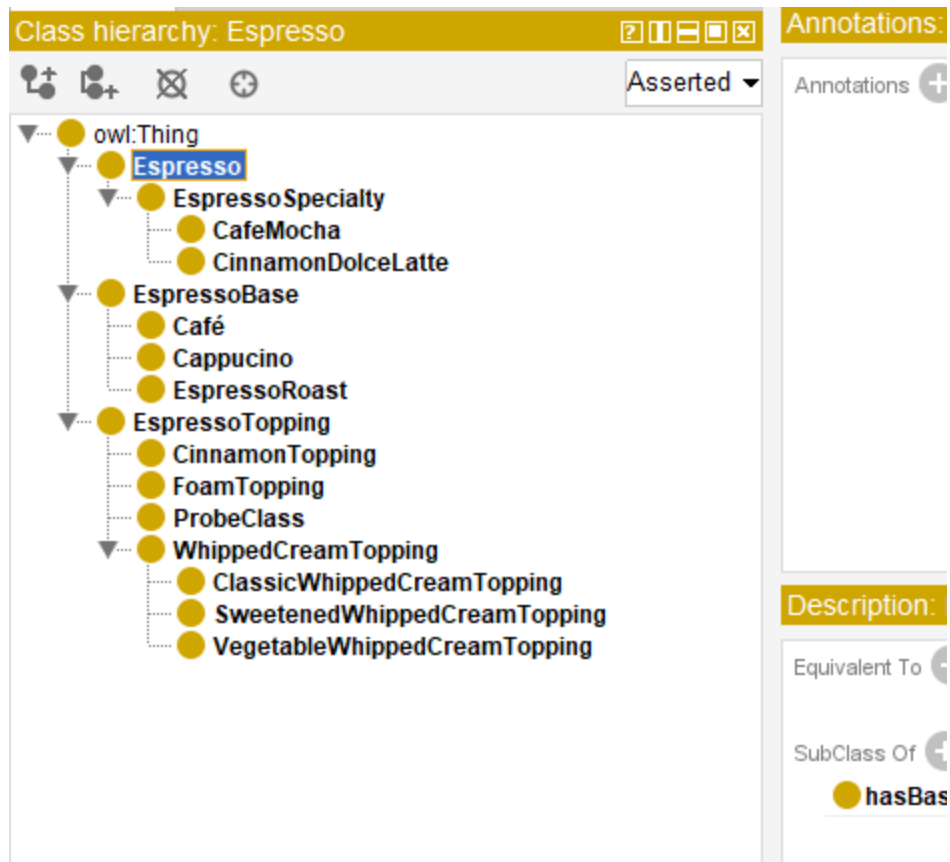
Exercise 19



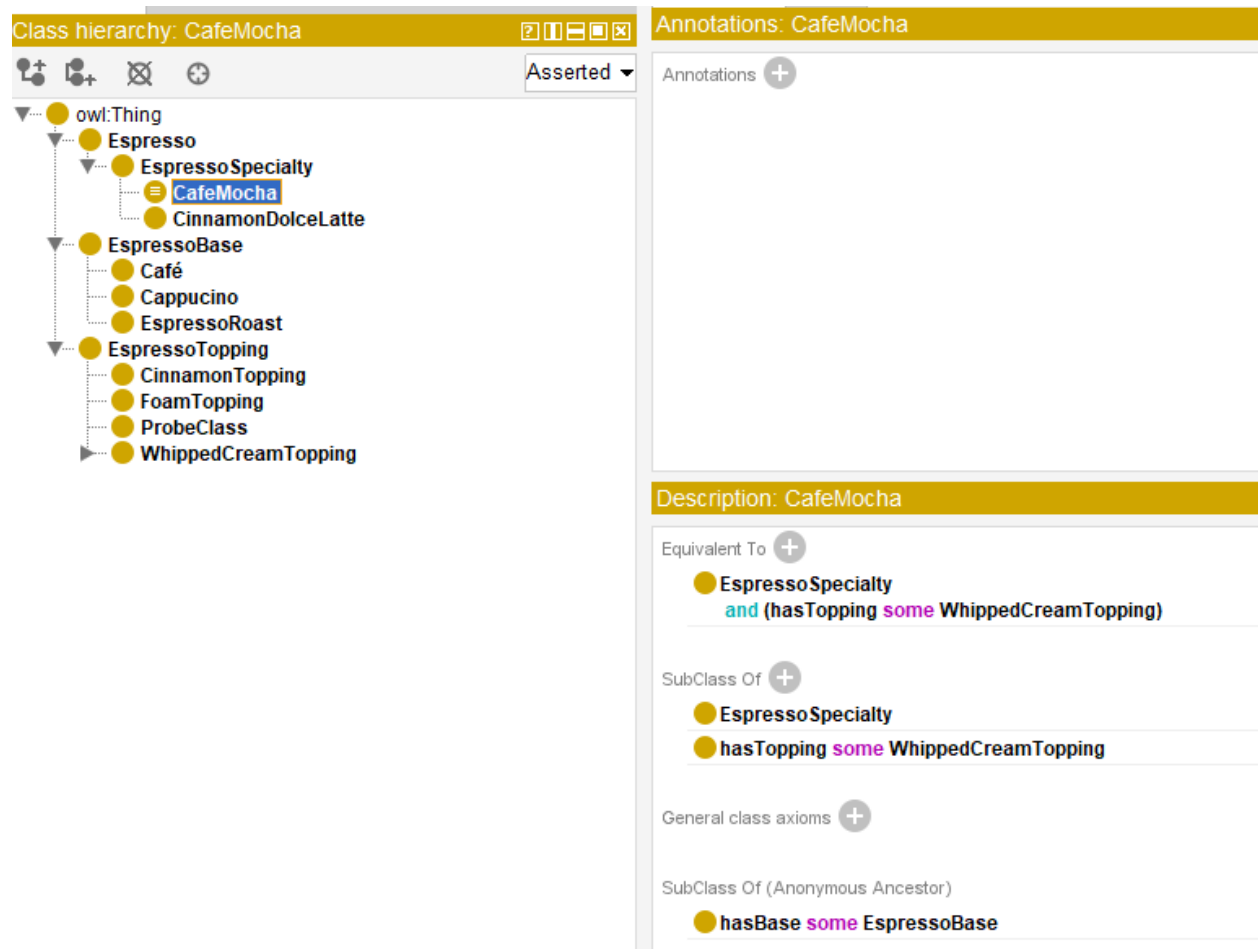
Exercise 20
Its not correct



Exercise 21



Exercise 22

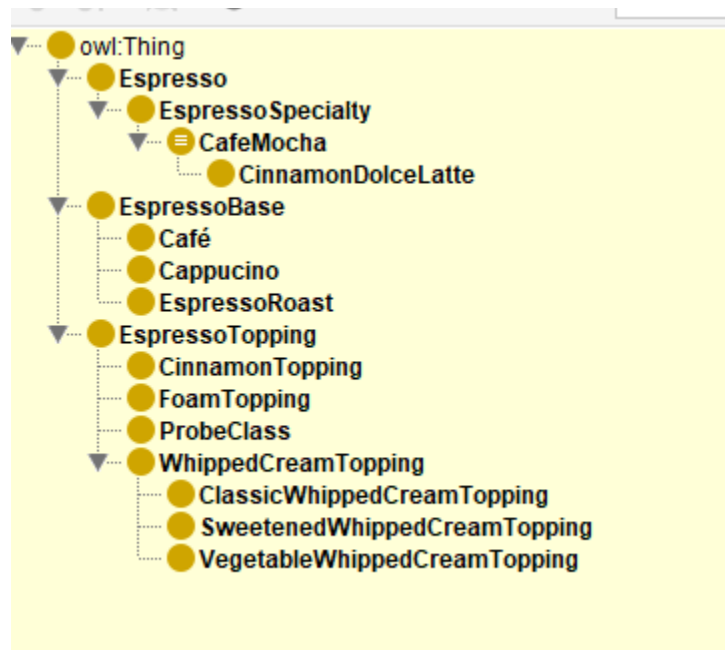
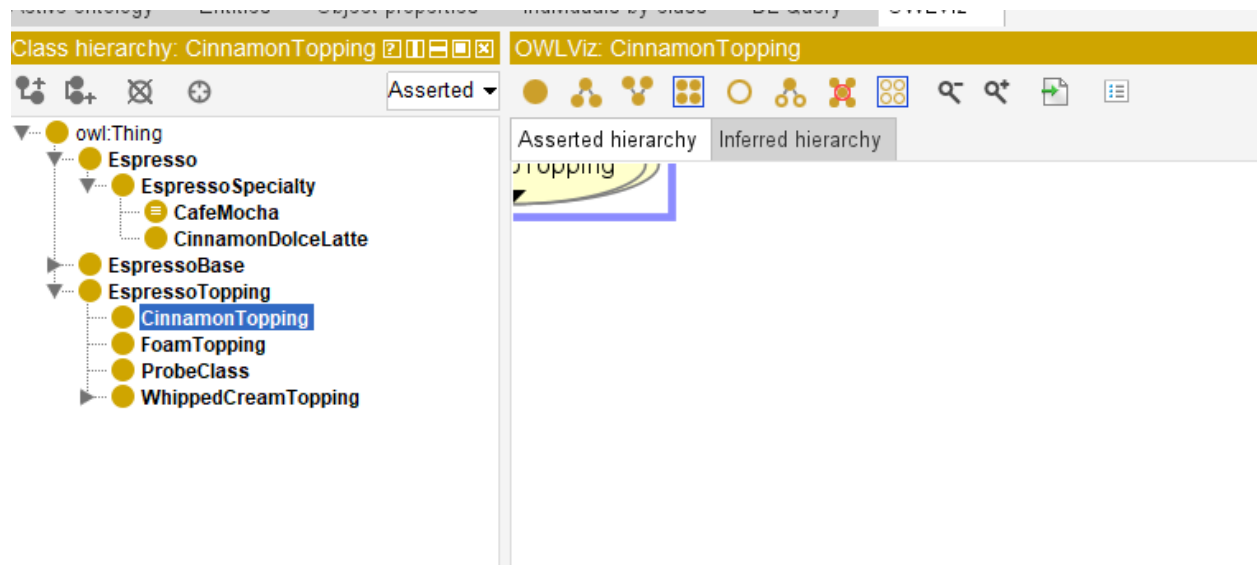


CafeMocha was converted to a defined class, the description changes from only SubClass Of conditions to an Equivalent To definition

Exercise 23

Cant really see anything, its bugged, also the answer to the question:

Classes that satisfy the definition of CaffeMocha can be automatically classified under CaffeMocha



Exercise 24

Classes

Object properties

Data properties

Class hierarchy: FoamOrClassicSpecialty

owl:Thing

Espresso

EspressoSpecialty

CafeMocha

CinnamonDolceLatte

FoamOrClassicSpecialty

EspressoBase

Café

Cappucino

EspressoRoast

EspressoTopping

CinnamonTopping

FoamTopping

ProbeClass

WhippedCreamTopping

ClassicWhippedCreamTopping

SweetenedWhippedCreamTopping

VegetableWhippedCreamTopping

Annotations

Usage

Annotations: FoamOrClassicSpecialty

Annotations

+

Description: FoamOrClassicSpecialty

Equivalent To

+

EspressoSpecialty

and (hasTopping only (ClassicWhippedCreamTopping or FoamTopping))

SubClass Of

+

EspressoSpecialty

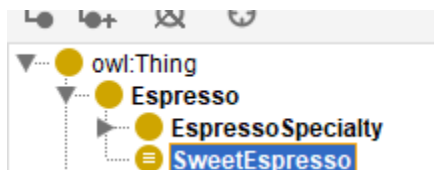
General class axioms

+

SubClass Of (Anonymous Ancestor)

hasBase some EspressoBase

Exercise 25



owl:topObjectProperty

- hasIngredient
- hasSweetness
- isBaseOf
- isIngredientOf
- isToppingOf

Characteristics

Description: hasSweetness

☒ Functional

☐ Inverse functional

☐ Transitive

☐ Symmetric

☐ Asymmetric

☐ Reflexive

☐ Irreflexive

Equivalent To

SubProperty Of

Inverse Of

Domains (intersection)

Ranges (intersection)

Disjoint With

SuperProperty Of (Chain)

EspressoSweetness

It should be functional because one espresso should have only a single sweetness level at a time

Classes

Object properties

Data properties

Class hierarchy: EspressoMachiato

owl:Thing

Espresso

EspressoSpecialty

CafeMocha

CinnamonDolceLatte

EspressoMachiato

FoamOrClassicSpecialty

SweetEspresso

EspressoBase

Café

Cappucino

EspressoRoast

EspressoSweetness

ExtraSweetness

MediumSweetness

NoneSweetness

EspressoTopping

CinnamonTopping

FoamTopping

ProbeClass

WhippedCreamTopping

Asserted

Annotations

Usage

Annotations: EspressoMachiato

Annotations

+

Description: EspressoMachiato

Equivalent To

+

SubClass Of

+

EspressoSpecialty

hasSweetness some ExtraSweetness

SweetEspresso

General class axioms

+

SubClass Of (Anonymous Ancestor)

hasBase some EspressoBase

Espresso

and (hasSweetness some ExtraSweetness)

Instances

+

Target for Key

+

Disjoint With

+

EspressoBase

EspressoTopping

The screenshot displays a web-based ontology editor interface. On the left, a 'Class hierarchy' panel shows a tree structure starting from 'owl:Thing'. Under 'Espresso', there is an 'EspressoSpecialty' class which includes 'CafeMocha', 'EspressoMachiato', and 'FoamOrClassicSpecialty'. Below 'EspressoSpecialty' is 'SweetEspresso', which in turn includes 'EspressoMachiato' (highlighted in blue). Other classes include 'EspressoBase' (with subclasses 'Café', 'Cappucino', 'EspressoRoast'), 'EspressoSweetness' (with subclasses 'ExtraSweetness', 'MediumSweetness', 'NoneSweetness'), and 'EspressoTopping' (with subclasses 'CinnamonTopping', 'FoamTopping', 'ProbeClass', 'WhippedCreamTopping').

On the right, the 'Annotations' tab is active for 'EspressoMachiato'. It shows a list of inferred properties and their values:

- Equivalent To:** A plus sign icon to expand the list.
- SubClass Of:** A plus sign icon to expand the list.
 - EspressoSpecialty**
 - hasSweetness** some **ExtraSweetness**
 - SweetEspresso**
- General class axioms:** A plus sign icon to expand the list.
- SubClass Of (Anonymous Ancestor):** A plus sign icon to expand the list.
 - hasBase** some **EspressoBase**
 - Espresso** and (hasSweetness some **ExtraSweetness**)
- Instances:** A plus sign icon to expand the list.
- Target for Key:** A plus sign icon to expand the list.
- Disjoint With:** A plus sign icon to expand the list.
 - EspressoBase**
 - EspressoTopping**

The reasoner inferred that espresso macchiato is a subclass of sweet espresso, because macchiato is an espresso which has extra sweetness

Exercise 26

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: InterestingEspresso

Annotations Usage

Annotations: InterestingEspresso

Annotations +

Description: InterestingEspresso

Equivalent To +

● Espresso
and (hasTopping min 3 EspressoTopping)

SubClass Of +

● Espresso

General class axioms +

SubClass Of (Anonymous Ancestor)

● hasBase some EspressoBase

owl:Thing

- Espresso
 - EspressoSpecialty
 - InterestingEspresso
 - SweetEspresso
 - EspressoBase
 - Café
 - Cappucino
 - EspressoRoast
 - EspressoSweetness
 - Extra Sweetness
 - Medium Sweetness
 - None Sweetness
 - EspressoTopping
 - CinnamonTopping
 - FoamTopping
 - ProbeClass
 - WhippedCreamTopping

Exercise 27

Data property hierarchy: hasCalories

owl:topDataProperty

hasCalories

Asserted

Annotations: hasCalories

Annotations

Characteristics: hasCalories

Functional

Equivalent To

SubProperty Of

Domains (intersection)

Ranges

Disjoint With

xsd:integer

Exercise 28

macchiato1

macchiato2

macchiato3

Description: macchiato1

Property assertions: macchiato1

Types

EspressoMachiato

Same Individual As

Different Individuals

Object property assertions

Data property assertions

hasCalories 80

Negative object property assertions

Negative data property assertions

Description: macchiato2

Property assertions: macchiato2

Types

EspressoMachiato

Same Individual As

Different Individuals

Object property assertions

Data property assertions

hasCalories 140

Negative object property assertions

Negative data property assertions

Description: :macchiato3

Property assertions: :macchiato3

Types

:EspressoMachiato

Same Individual As

Different Individuals

Object property assertions

Data property assertions

:hasCalories 260

Negative object property assertions

Negative data property assertions

Exercise 29

Description: :macchiato1

Property assertions: :macchiato1

Types

:EspressoMachiato

:LowCalorieEspresso

Same Individual As

Different Individuals

Object property assertions

Data property assertions

:hasCalories 80

Negative object property assertions

Negative data property assertions

Description: :macchiato2

Types

:EspressoMachiato

?

@

x

o

:MediumCalorieEspresso

?

@

Same Individual As

Different Individuals

Property assertions: :macchiato2

Object property assertions

Data property assertions

:hasCalories

140

?

@

x

o

Negative object property assertions

Negative data property assertions

Description: :macchiato3

Types

:EspressoMachiato

?

@

x

o

:HighCalorieEspresso

?

@

Same Individual As

Different Individuals

Property assertions: :macchiato3

Object property assertions

Data property assertions

:hasCalories

260

?

@

x

o

Negative object property assertions

Negative data property assertions